SULIT

UTM UNIVERSITI TEKNOLOGI MALAYSIA

Fakulti
Sains Komputer
& Sistem Maklumat

UNIVERSITI PENYELIDIKAN

PEPERIKSAAN AKHIR SEMESTER I 2012/2013

KOD KURSUS	**: SCI 2143**
NAMA KURSUS	**:PROBABILITY & STATISTICAL DATA ANALYSIS**
TAHUN/PROGRAM	**:**
MASA	**: 3 HOURS**
TARIKH	**:**
TEMPAT	**:**
(ARAHAN KEPADA CALON)	**:**

Answer all questions in the question booklet provided.

(sila catatkan nama pensyarah dan seksyen di dalam kertas jawapan anda)

Nama	
No. K/P	
Tahun/Program	
Seksyen	
Nama Pensyarah	

KERTAS INI TERDIRI DARIPADA .....7..... MUKA SURAT SAHAJA TIDAK TERMASUK MUKA SURAT INI.

Question 1 **(10 marks)**

(i) Express the null hypothesis H_0 and alternative hypothesis H_1 in symbolic form for the given statement. Use the correct symbol (μ, σ, p) for appropriate indicated parameter.

a. The mean salary of statistics professors is at least RM25000. (2 marks)

b. The proportion of defective computers is less than 0.05. (2 marks)

c. Salaries among women business analysts have a standard deviation greater than RM3000. (2 marks)

(ii) Give the definition of

a. Type I error. (2 marks)

b. Type II error. (2 marks)

Question 2 **(20 marks)**

(i) In previous tests, baseballs were dropped 24ft onto a concrete surface, and they bounced an average of 92.84 inch. In a test of a sample of 40 new balls, the bounce heights had a mean of 92.67 inch and a standard deviation of 1.79 inch. Use a 0.05 significance level to determine whether there is sufficient evidence to support the claim that the new balls have bounce heights with a mean different from 92.84 inch. (5 marks)

(ii) Because of the expense involved, car crash tests often use small samples. When five BMW cars are crashed under standard conditions, the repair costs (in dollars) are as shown in the accompanying list. Use a 0.05 significance level to test the claim that the mean for all BMW cars is less than $1000. (5 marks)

797 571 904 1147 418

(iii) Use a 0.01 significance level to test the claim that the two samples comes from populations with the same mean. Assume that the two samples are independent simple random samples. (5 marks)

Table 1

Treatment Group	Placebo Group
$n_1 = 50$	$n_2 = 100$
$\bar{x}_1 = 7.00$	$\bar{x}_2 = 6.00$
$s_1 = 1.00$	$s_2 = 2.00$

(iv) Table 2 shows the measured nicotine contents of randomly selected filtered and non-filtered king-size cigarettes. All measurements are in milligrams. Use a 0.05 significance level to test the claim that the mean amount of nicotine in filtered king-size cigarettes is equal to the mean amount of nicotine in non-filtered king-size cigarettes. (5 marks)

Table 2

Filtered	Non-filtered
$n_1 = 21$	$n_2 = 8$
$\bar{x}_1 = 0.94$	$\bar{x}_2 = 1.65$
$s_1 = 0.31$	$s_2 = 0.16$

Formula:

$$\text{standard deviation, } s = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}}$$

$$z_0 = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \qquad t_0 = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$$

$$z_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \qquad S_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$$

$$S_p = \sqrt{S_p^2}$$

$$t_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Question 3 **(20 marks)**

(i) Given the following 5 pairs of data set in Table 3, find

Table 3

x	1	2	3	5	5
y	19	12	4	20	18

(a) SS_{xy} (3 marks)

(b) SS_{xx} (2 marks)

(c) SS_{yy} (2 marks)

(d) Correlation coefficient, r (2 marks)

(e) Based on answer in (d), what can you conclude about the relationship between x and y? (1 mark)

(ii) Raw material used in the production of a synthetic fibre is stored in a place which has no humidity control. Measurements of the relative humidity in the storage place and the moisture content of a sample of the raw material (both in percentages) on 12 days yielded the following results in Table 4.

Table 4

Humidity, x	Moisture content, y
42	12
35	8
50	14
43	9
48	11
62	16

Humidity, x	Moisture content, y
31	7
36	9
44	12
39	10
55	13
48	11

(a) Calculate SS_{xx} and SS_{yy}. (4 marks)

(b) Calculate value b and a. (2 marks)

(c) Based on answers in (a) and (b), find the least square regression line $\hat{y} = a + bx$ for the data on humidity and moisture content of the 12 samples of the raw materials. (2 marks)

(d) Predict the moisture content of the raw material when the humidity of the storage place is 43.5%. (2 marks)

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Formula:

$$SS_{xy} = \sum xy - \frac{\sum x \sum y}{n} \qquad SS_{xx} = \sum x^2 - \frac{(\sum x)^2}{n}$$

$$SS_{yy} = \sum y^2 - \frac{(\sum y)^2}{n} \qquad r = \frac{SS_{xy}}{\sqrt{SS_{xx} SS_{xy}}}$$

Question 4 **(10 marks)**

A recycling company collects the waste discarded by households in a region. Discarded waste must be separated into categories of metal, paper, plastic and glass. In planning for the equipment needed to collect and process the garbage, the company needs to know the weights (in kg) for each category, as summarized in Table 5. The company wants to test the claim that the four specific populations of garbage have the same mean.

Table 5

Metal	Paper	Plastic	Glass
$n = 62$	$n = 62$	$n = 62$	$n = 62$
$\bar{x} = 2.218$	$\bar{x} = 9.428$	$\bar{x} = 1.911$	$\bar{x} = 3.752$
$s = 1.091$	$s = 4.168$	$s = 1.065$	$s = 3.108$

(i) State the null hypothesis and alternative hypothesis. (2 marks)

(ii) Find variance between sample, $ns_{\bar{x}}^2$. (2 marks)

(iii) Find variance within sample, s_p^2. (2 marks)

(iv) Calculate F test statistic based on answers in (ii) and (iii). (1 mark)

(v) Determine the numerator and the denominator. (1 mark)

(vi) Find F critical value at 0.05 significance level based on answers in (v). (1 mark)

(vii) At the 0.05 significance level, what do you conclude about the claim that the four specific populations of garbage have the same mean? (1 mark)

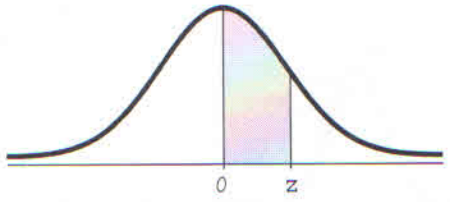


TABLE A-2 Standard Normal (z) Distribution

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495 *	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949 *	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.10 and higher	.4999									

NOTE: For values of z above 3.09, use 0.4999 for the area.

*Use these common values that result from interpolation:

z score	Area
1.645	0.4500
2.575	0.4950

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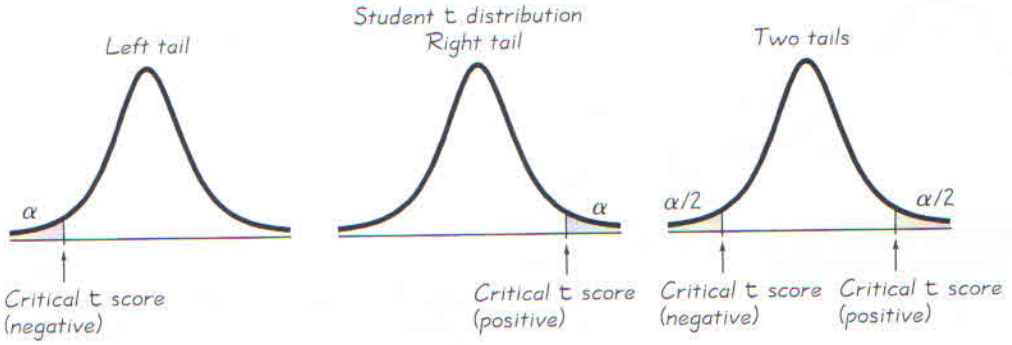
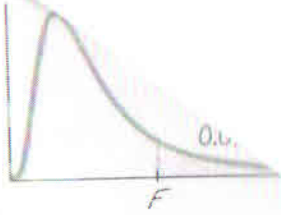


TABLE A-3 t Distribution

Degrees of Freedom	α: .005 (one tail) .01 (two tails)	.01 (one tail) .02 (two tails)	.025 (one tail) .05 (two tails)	.05 (one tail) .10 (two tails)	.10 (one tail) .20 (two tails)	.25 (one tail) .50 (two tails)
1	63.657	31.821	12.706	6.314	3.078	1.000
2	9.925	6.965	4.303	2.920	1.886	.816
3	5.841	4.541	3.182	2.353	1.638	.765
4	4.604	3.747	2.776	2.132	1.533	.741
5	4.032	3.365	2.571	2.015	1.476	.727
6	3.707	3.143	2.447	1.943	1.440	.718
7	3.500	2.998	2.365	1.895	1.415	.711
8	3.355	2.896	2.306	1.860	1.397	.706
9	3.250	2.821	2.262	1.833	1.383	.703
10	3.169	2.764	2.228	1.812	1.372	.700
11	3.106	2.718	2.201	1.796	1.363	.697
12	3.054	2.681	2.179	1.782	1.356	.696
13	3.012	2.650	2.160	1.771	1.350	.694
14	2.977	2.625	2.145	1.761	1.345	.692
15	2.947	2.602	2.132	1.753	1.341	.691
16	2.921	2.584	2.120	1.746	1.337	.690
17	2.898	2.567	2.110	1.740	1.333	.689
18	2.878	2.552	2.101	1.734	1.330	.688
19	2.861	2.540	2.093	1.729	1.328	.688
20	2.845	2.528	2.086	1.725	1.325	.687
21	2.831	2.518	2.080	1.721	1.323	.686
22	2.819	2.508	2.074	1.717	1.321	.686
23	2.807	2.500	2.069	1.714	1.320	.685
24	2.797	2.492	2.064	1.711	1.318	.685
25	2.787	2.485	2.060	1.708	1.316	.684
26	2.779	2.479	2.056	1.706	1.315	.684
27	2.771	2.473	2.052	1.703	1.314	.684
28	2.763	2.467	2.048	1.701	1.313	.683
29	2.756	2.462	2.045	1.699	1.311	.683
Large (z)	2.575	2.326	1.960	1.645	1.282	.675

TABLE A-5 *F* Distribution ($\alpha = 0.05$ in the right tail)



Denominator degrees of freedom (df_2)	Numerator degrees of freedom (df_1)								
	1	2	3	4	5	6	7	8	9
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54
2	18.513	19.000	19.164	19.247	19.296	19.330	19.353	19.371	19.385
3	10.128	9.5521	9.2766	9.1172	9.0135	8.9406	8.8867	8.8452	8.8123
4	7.7086	6.9443	6.5914	6.3882	6.2561	6.1631	6.0942	6.0410	6.9988
5	6.6079	5.7861	5.4095	5.1922	5.0503	4.9503	4.8759	4.8183	4.7725
6	5.9874	5.1433	4.7571	4.5337	4.3874	4.2839	4.2067	4.1468	4.0990
7	5.5914	4.7374	4.3468	4.1203	3.9715	3.8660	3.7870	3.7257	3.6767
8	5.3177	4.4590	4.0662	3.8379	3.6875	3.5806	3.5005	3.4381	3.3881
9	5.1174	4.2565	3.8625	3.6331	3.4817	3.3738	3.2927	3.2296	3.1789
10	4.9646	4.1028	3.7083	3.4780	3.3258	3.2172	3.1355	3.0717	3.0204
11	4.8443	3.9823	3.5874	3.3567	3.2039	3.0946	3.0123	2.9480	2.8962
12	4.7472	3.8853	3.4903	3.2592	3.1059	2.9961	2.9134	2.8486	2.7964
13	4.6672	3.8056	3.4105	3.1791	3.0254	2.9153	2.8321	2.7669	2.7144
14	4.6001	3.7389	3.3439	3.1122	2.9582	2.8477	2.7642	2.6987	2.6458
15	4.5431	3.6823	3.2874	3.0556	2.9013	2.7905	2.7066	2.6408	2.5876
16	4.4940	3.6337	3.2389	3.0069	2.8524	2.7413	2.6572	2.5911	2.5377
17	4.4513	3.5915	3.1968	2.9647	2.8100	2.6987	2.6143	2.5480	2.4943
18	4.4139	3.5546	3.1599	2.9277	2.7729	2.6613	2.5767	2.5102	2.4563
19	4.3807	3.5219	3.1274	2.8951	2.7401	2.6283	2.5435	2.4768	2.4227
20	4.3512	3.4928	3.0984	2.8661	2.7109	2.5990	2.5140	2.4471	2.3928
21	4.3248	3.4668	3.0725	2.8401	2.6848	2.5727	2.4876	2.4205	2.3660
22	4.3009	3.4434	3.0491	2.8167	2.6613	2.5491	2.4638	2.3965	2.3419
23	4.2793	3.4221	3.0280	2.7955	2.6400	2.5277	2.4422	2.3748	2.3201
24	4.2597	3.4028	3.0088	2.7763	2.6207	2.5082	2.4226	2.3551	2.3002
25	4.2417	3.3852	2.9912	2.7587	2.6030	2.4904	2.4047	2.3371	2.2821
26	4.2252	3.3690	2.9752	2.7426	2.5868	2.4741	2.3883	2.3205	2.2655
27	4.2100	3.3541	2.9604	2.7278	2.5719	2.4591	2.3732	2.3053	2.2501
28	4.1960	3.3404	2.9467	2.7141	2.5581	2.4453	2.3593	2.2913	2.2360
29	4.1830	3.3277	2.9340	2.7014	2.5454	2.4324	2.3463	2.2783	2.2229
30	4.1709	3.3158	2.9223	2.6896	2.5336	2.4205	2.3343	2.2662	2.2107
40	4.0847	3.2317	2.8387	2.6060	2.4495	2.3359	2.2490	2.1802	2.1240
60	4.0012	3.1504	2.7581	2.5252	2.3683	2.2541	2.1665	2.0970	2.0401
120	3.9201	3.0718	2.6802	2.4472	2.2899	2.1750	2.0868	2.0164	1.9588
∞	3.8415	2.9957	2.6049	2.3719	2.2141	2.0986	2.0096	1.9384	1.8799